

Geometry
Unit 2
Lesson 7 Homework

Name _____
Date _____
Period _____

Complete the proof.

Given: $AB = CD$

Prove: $AC = BD$



	Statement	Reason
1.	$AB = CD$	
2.		Reflexive Property of Equality
3.	$AB + BC = BC + CD$	
4.		Segment Addition Postulate
5.		Substitution Property of Equality

Complete the paragraph proof.

Given: $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are complementary.

Prove: $\angle 1 \cong \angle 3$

It is 6. _____ that $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are

complementary. By the definition of 7. _____,

$m\angle 1 + m\angle 2 = 90^\circ$ and $m\angle 2 + m\angle 3 = 90^\circ$. By the 8. _____ property,

$m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$.

Then, $m\angle 1 =$ 9. _____ by the subtraction property. Therefore, $\angle 1 \cong \angle 3$ by the

10. _____.